

Abstract No.

## Transcriptomics and proteomics pave the way to the identification of novel regulators of plasticity in Glioblastoma

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Glioblastoma (GBM) is a highly aggressive brain tumor, strongly resistant to therapy and associated with poor survival. Increasing evidence suggests that phenotypic plasticity, defined as the ability of tumor subpopulations to switch between distinct cellular states, contributes to therapy escape. Although the coexistence of multiple GBM states is well established, the mechanisms sustaining their plasticity remain unclear.

Here, we investigated intra-tumoral heterogeneity and plasticity in 3D GBM models. Transcriptional profiles of primary GBM samples revealed marked heterogeneity, consistent with the coexistence of distinct cell states. Based on our data and previous observations, we selected CD44 and CD24 cell-surface markers to identify and isolate subpopulations from a GBM sample (CL\_L0627). Notably, only the CD24-negative subpopulations were able to reconstitute the parental population, which we therefore defined as plastic. To assess whether this phenotype was present also in vivo, we performed orthotopic xenograft experiments. Tumors derived from CD24-negative cells displayed increased growth compared with those generated from CD24-positive cells, suggesting that the plastic subpopulation is a major driver of tumor expansion. To uncover the mechanisms underlying plasticity in vitro, we integrated bulk-RNAseq and proteomics analyses. Plastic cells showed selective enrichment of mitochondria-related pathways and increased mitochondrial ROS levels, without changes in mitochondrial mass, indicating that metabolic rewiring may support proliferation and cell-state transitions. Conversely, non-plastic cells were enriched for glycogen metabolism and extracellular vesicle-related pathways, suggesting a possible metabolic and paracrine supportive role. Finally, transcriptomic analyses determined an enrichment of Polycomb repressive complex 2 (PRC2) genes, with intriguingly distinct signatures in plastic and non-plastic cells. These results suggest that Polycomb may act as an upstream regulator of alternative cellular states, orchestrating differential epigenetic landscapes consistent with its role in cellular identity. Future studies will determine whether PRC2-dependent chromatin regulation and mitochondrial rewiring cooperate to sustain GBM plasticity and aggressiveness.



Supported by SciSerTec

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